RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Tuesday, 15 September 2026, 12:13 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 12:15 AM
Time taken	1 min 3 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Function to find the total ways using O(1) auxiliary space
long long countWays(int n) {
    if (n < 0) return 0;
    if (n == 0 || n == 1 || n == 2) return 1;

    // Base cases for index 0, 1, and 2
    long long ways0 = 1; // ways(n-3) initially for n=3
    long long ways1 = 1; // ways(n-2) initially for n=3
    long long ways2 = 1; // ways(n-1) initially for n=3
    long long current_ways = 0;

    for (int i = 3; i <= n; i++) {
        current_ways = ways2 + ways0; // ways(i-1) + ways(i-3)

        // Shift values forward for the next iteration
        ways0 = ways1;
        ways1 = ways2;
        ways2 = current_ways;
    }

    return current_ways;
}

int main() {
    int n;

    // Read input
    if (scanf("%d", &n) == 1) {
        // Output the result
        printf("%lld\n", countWays(n));
    }
```

```
34
35      return 0;
36  }
37
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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